

Risk-Budgeted Imitative Reinforcement Learning for Multi-Level Market Making

Inventory-risk control through explicit constraints

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Talk Outline

This shortened version covers the first part of the talk: from the limit order book to the unconstrained IMM agent. Selected references are kept at the end.

Market-making setup

1. **Limit order book**
prices, queue sizes, and order flow
2. **Liquidity provision**
spread capture creates inventory risk
3. **Sequential decision problem**
MDP, reward, value, and Bellman relation

From PPO to IMM

4. **Previous PPO baseline**
fixed catalogue of quote configurations
5. **Continuous multi-level actions**
spreads, sizes, and reservation-price shift
6. **IMM with TD3**
predictive representation and imitation loss

Limit Order Books: Prices, Volumes, and Order Flow

At each decision time the policy sees a finite-depth snapshot of buy and sell liquidity.

$$L_t = \left(\{(p_{t,i}^b, v_{t,i}^b)\}_{i=1}^D, \{(p_{t,i}^a, v_{t,i}^a)\}_{i=1}^D \right)$$

Best quotes and spread

$$m_t = \frac{p_{t,1}^a + p_{t,1}^b}{2}, \quad s_t = p_{t,1}^a - p_{t,1}^b.$$

Book event

$$L_{t+1} = \mathcal{T}(L_t, \varepsilon_{t+1}),$$

ε_{t+1} collects submissions, executions, and cancellations.

Qualification The observation is a finite-depth, discrete-time projection of the market, not its complete state.

Market Making as Liquidity Provision

A market maker posts both sides of the book. Profit appears only if executions are balanced.

$$\tilde{p}_t^b = m_t - \delta_t^b, \quad \tilde{p}_t^a = m_t + \delta_t^a$$

Both sides fill

$$R_t^{\text{spread}} = u(\delta_t^a + \delta_t^b) > 0.$$

Spread is captured.

Only one side fills

$$z_{t+1} = z_t + e_t^b - e_t^a \neq 0.$$

Inventory remains exposed.

Key point The action affects immediate spread revenue and the risk carried into future decisions.

Avellaneda and Stoikov (2008), *High-frequency trading in a limit order book*.

Inventory Risk and Mark-to-Market Wealth

Mark-to-market wealth separates earned spread from the price risk of the remaining position.

$$W_t = x_t + z_t m_t, \quad \text{PnL}_{0:T} = W_T - W_0$$

One-step decomposition

$$\Delta W_t = \underbrace{\delta_t^b e_t^b + \delta_t^a e_t^a}_{\text{spread capture}} + \underbrace{z_{t+1} \Delta m_t}_{\text{inventory revaluation}} - \underbrace{f_t}_{\text{fees}}$$

Key point Inventory risk enters through $z_{t+1} \Delta m_t$: the same price move matters more when the position is larger.

Market Making as a Sequential Decision Process

A quote affects both current spread revenue and the inventory carried into future decisions.

$$\mathcal{M} = (\mathcal{X}, \mathcal{A}, P, r, \gamma)$$

$$X_t \xrightarrow{O} S_t \xrightarrow{\pi_\theta} a_t, \quad (r_t, X_{t+1}) \sim P(\cdot \mid X_t, a_t)$$

Policy view

$S_t = O(X_t)$ is the information available to the agent; the policy chooses a_t from this observation.

Finite-horizon objective

$$G_t = \sum_{k=0}^{T-t-1} \gamma^k r_{t+k}, \quad \max_{\theta} J(\pi_\theta) = \mathbb{E}[G_0].$$

Key point RL optimizes cumulative return rather than immediate spread revenue.

Simulator State vs. Policy Observation

The simulator has a Markov internal state, while the neural policy acts only on an observation of that state.

Simulator state X_t

$X_t = (\text{replay index}, L_t, z_t, x_t, \text{orders}, \text{fills})$

Contains the information needed by the replay environment to generate the next transition.

Markov by construction.

Markov condition: $\Pr(r_t, X_{t+1} \mid \mathcal{H}_t, a_t) = \Pr(r_t, X_{t+1} \mid X_t, a_t).$

Policy observation S_t

$S_t = O(X_t) = \text{Enc}(H_t, \xi_t, u_t)$

H_t : recent LOB history; ξ_t : market signals;
 u_t : agent private state.

Generally $S_t \neq X_t$.

Qualification Bellman value functions are defined on the Markov simulator state X_t ; the neural policy itself faces partial observability through S_t .

Value Functions and the Bellman Relation

The Bellman relation is written for the Markov simulator state X_t ; the policy observes $S_t = O(X_t)$.

$$\begin{aligned} Q_t^\pi(x, a) &= \mathbb{E}_\pi[G_t \mid X_t = x, A_t = a], \\ &= \mathbb{E}[r_t + \gamma V_{t+1}^\pi(X_{t+1}) \mid X_t = x, A_t = a] \end{aligned}$$

State value

$$V_t^\pi(x) = \mathbb{E}_{a \sim \pi_t(\cdot | O(x))} Q_t^\pi(x, a)$$

Advantage

$$A_t^\pi(x, a) = Q_t^\pi(x, a) - V_t^\pi(x)$$

Key point For a fixed policy, V^π averages Q^π . The maximum appears only for the optimal value: $V_t^*(x) = \max_a Q_t^*(x, a)$.

Previous Study: A Discrete Market-Making Agent

The previous agent compressed recent book history and selected one of nine categorical actions.

$$h_t = \left[f_{\psi}(\Phi(L_{t-K+1:t})), z_t, \tau_t \right]$$

Categorical actor

$$\pi_{\theta}(\cdot \mid S_t) = \text{softmax}(g_{\theta}(h_t)).$$

State-value critic

$$V_{\omega}(S_t) = v_{\omega}(h_t).$$

Key point The CNN representation is shared; the actor chooses a category and the critic estimates future return.

Previous Objective: Reward-Based Inventory Control

The previous agent controlled inventory indirectly by penalizing it inside the training reward.

$$r_t^{\text{old}} = r_t^{\text{econ}} - \eta \ell(z_t)$$

$$\pi_\eta \in \arg \max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^{T-1} \gamma^t r_t^{\text{old}} \right]$$

What η controls

A larger η puts stronger pressure on the policy to avoid large inventory.

What η does not specify

The achieved exposure is only observed after training:

$$\eta \longrightarrow \pi_\eta \longrightarrow J_C(\pi_\eta).$$

Key point The penalty controls the strength of the preference against inventory, not the risk level achieved ex ante.

Discrete Quoting: A Fixed Catalogue of Actions

The policy did not choose prices and sizes independently; it selected a predefined template.

$$\mathcal{A}_{\text{old}} = \{\text{cancel_all}\} \cup \{(\kappa_j^b, \kappa_j^a, u_j^b, u_j^a)\}_{j=1}^8$$

Template j becomes executable quotes

$$\tilde{p}_t^b = p_{t,1}^b - \kappa_j^b \Delta, \quad \tilde{p}_t^a = p_{t,1}^a + \kappa_j^a \Delta.$$

Qualification The catalogue contains no bid-only or ask-only action, so direct one-sided inventory correction is unavailable.

PPO: Three Components of the Training Objective

$$\max_{\theta, \omega} \mathcal{L}^{\text{PPO}} = \underbrace{L^{\text{CLIP}}(\theta)}_{\text{actor}} - \underbrace{c_V L_V(\omega)}_{\text{critic}} + \underbrace{c_H \hat{H}(\pi_\theta)}_{\text{exploration}}$$

1. Actor — policy improvement

$$\varrho_t = \frac{\pi_\theta(a_t | S_t)}{\pi_{\theta_{\text{old}}}(a_t | S_t)}$$

$$L^{\text{CLIP}} = \hat{\mathbb{E}}_t \min \left\{ \varrho_t \hat{A}_t, \text{clip}(\varrho_t, 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right\}.$$

2. Critic — value regression

$$\hat{V}_t^{\text{target}} = V_{\omega_{\text{old}}}(S_t) + \hat{A}_t, \quad L_V(\omega) = \hat{\mathbb{E}}_t \left[\left(V_\omega(S_t) - \hat{V}_t^{\text{target}} \right)^2 \right].$$

3. Exploration — entropy bonus

$$\hat{H} = -\hat{\mathbb{E}}_t \sum_a \pi_\theta(a | S_t) \log \pi_\theta(a | S_t)$$

Previous Results: Active Quoting and Downside Exposure

The previous PPO agent traded actively, but its PnL distribution retained a substantial negative tail.

Validation: 20 runs $\times$ 20,000 steps

Metric	Reported value
Median PnL	+0.309 USDT
Mean PnL	+0.115 USDT
5% PnL quantile	−1.190 USDT
Quoted rate	0.873

Key point Quoted rate is **87.3%**, yet the lower 5% PnL quantile is **−1.190 USDT**. Activity alone does not control downside exposure.

What the Previous Formulation Leaves Unresolved

Three design choices prevent the old policy from targeting a specified inventory-risk level.

Restriction	Formal consequence
Fixed symmetric catalogue	$\mathcal{A}_{\text{agent}} \subsetneq \mathcal{A}_{\text{baseline}}$
Reward-only risk treatment	η specified, but $J_C(\pi_\eta)$ unknown ex ante
No explicit admissible set	$\Pi_B = \{\pi : J_C(\pi) \leq B\}$ not defined

$$\underbrace{\text{continuous multi-level actions}}_{\text{richer control}} + \underbrace{J_C(\pi) \leq B}_{\text{explicit risk target}}$$

From Discrete Quoting to Risk-Budgeted Market Making

The new study expands the action space and replaces an implicit risk preference with an explicit constraint.

Dimension	Previous study	Current research
Actions	Fixed discrete configurations	Continuous parameters for multi-level quotes
Learning	CNN-PPO	TD3-based imitative learning
Guidance	None	Signal-based expert actions
Risk control	Reward coefficient η	Constraint $J_C(\pi) \leq B$
Estimand	Quoting behavior	Risk response and economic trade-off

Current Problem: Sequential Multi-Level Quoting

The policy produces a structured order set; fills update cash and inventory before the next decision.

$$S_t \xrightarrow{\pi_\theta} a_t \xrightarrow{\text{Decode}} \mathcal{O}_t \xrightarrow{\mathcal{F}} (e_t^b, e_t^a) \xrightarrow{\text{accounting}} (z_{t+1}, x_{t+1})$$

Two outcomes, two objectives

$$(r_t, c_t) = \left(r_t^{\text{econ}}, \left| \frac{z_t}{N_a} \right| \right)$$

Key point Return and inventory exposure are measured separately; the constraint does not hide risk inside the reward.

Historical Replay and Execution Assumptions

Historical replay fixes the external market path; the agent affects only its own hypothetical orders and accounting.

$$P(M_{t+1} \mid \mathcal{H}_t, a_t) = P(M_{t+1} \mid \mathcal{H}_t)$$

Executions are simulated conditionally on the replay

$$(e_t^b, e_t^a) \sim K_{\mathcal{F}}(\cdot \mid \mathcal{O}_t, M_{t:t+1}, U_t), \quad U_{t+1} = g(U_t, \mathcal{O}_t, e_t^b, e_t^a).$$

Qualification All PnL conclusions are conditional on zero market impact and the chosen queue, latency, and fill assumptions.

Market Information and Private State

The policy state combines recent market dynamics with signals and agent-specific information.

$$S_t = \text{Enc}(H_t, \xi_t, u_t), \quad H_t = (L_{t-63}, \dots, L_t)$$

Component	Dimension / horizon	Information
LOB history H_t	depth 5, lookback 64	prices, volumes, book dynamics
Signals ξ_t	$h \in \{10, 60, 120, 300\}$ s	predictive market features
Private state u_t	current time t	inventory and active orders

Key point Every input to S_t must be measurable at decision time t ; future targets may train the encoder but never enter the policy input.

Continuous Parameters for Multi-Level Orders

Four continuous policy outputs are decoded into prices and sizes at several book levels.

$$a_t = (a_{t,1}, \dots, a_{t,4}) \in \mathcal{A} \subseteq \mathbb{R}^4, \quad \mathcal{O}_t = \text{Decode}(S_t, a_t)$$

Structured order collection

$$\begin{aligned} \mathcal{O}_t &= \{(p_{t,\ell}^s, u_{t,\ell}^s) : s \in \{b, a\}, \ell = 1, \dots, M\}, \\ p_{t,\ell}^b &\leq p_{t,1}^b < p_{t,1}^a \leq p_{t,\ell}^a, \quad u_{t,\ell}^s \geq 0. \end{aligned}$$

Key point The decoder guarantees valid quote geometry; its image defines which multi-level orders the policy can express.

Multi-level action representation: Niu et al. (2023), IMM. Action dimension refers to the current research setup.

Inventory Exposure as a Separate Cost

Risk is the expected discounted magnitude of normalized inventory, kept separate from economic return.

$$c_t = \left| \frac{z_t}{N_a} \right|, \quad C_i = (1 - \gamma_C) \sum_{t=0}^{T-1} \gamma_C^t c_{i,t}, \quad J_C(\pi) = \mathbb{E}_\pi[C_i]$$

Current horizon and normalization

$$\gamma_C = 0.999, \quad T = 3600, \quad \sum_{t=0}^{T-1} (1 - \gamma_C) \gamma_C^t = 1 - \gamma_C^T \approx 0.973.$$

Qualification $J_C(\pi) \leq B$ constrains expected exposure, not the inventory of every episode or every time step.

Classical Benchmarks for Inventory-Aware Quoting

Baselines separate inactivity, fixed quoting, heuristic inventory control, and model-based control.

Benchmark	Control role	Formula / implementation
No Trade	no active orders	$u_t^b = u_t^a = 0$
Fixed Quote	constant distances and sizes	$(\delta_t^b, \delta_t^a, u_t^b, u_t^a) = \text{const}$
Inventory-Skew	shift quote center against inventory	center $m_t - g(z_t)$
Avellaneda–Stoikov	reservation price, spread and reference quantities	$p_t^{\text{res}} = m_t - z_t \rho \sigma^2 (T - t),$ $\Delta_t^* = \rho \sigma^2 (T - t) + \frac{2}{\rho} \ln \left(1 + \frac{\rho}{k} \right),$ $p_t^b = p_t^{\text{res}} - \Delta_t^* / 2, \quad p_t^a = p_t^{\text{res}} + \Delta_t^* / 2,$ $u_t^b = q_t^{b, \text{ref}}, \quad u_t^a = q_t^{a, \text{ref}}.$

Avellaneda and Stoikov (2008), *High-frequency trading in a limit order book*.

Baseline Taxonomy and Experimental Roles

Each comparison answers a different question; only matched branches identify the risk mechanism.

Comparison block	Methods
Market controls	No Trade, Fixed Quote, Inventory-Skew, Avellaneda–Stoikov
Learning controls	LTIIC expert, TD3, IMM
Risk controls	Penalty-IMM (η), RB-IMM (B)

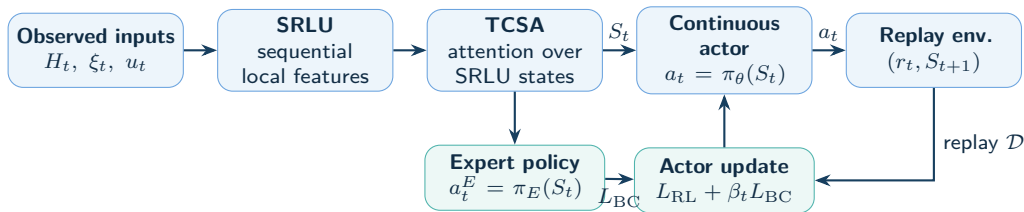
imitation effect : IMM – TD3,

risk mechanism : RB-IMM(B) – Penalty-IMM(η) at matched J_C

Imitative learning reference: Niu et al. (2023), IMM; TD3: Fujimoto et al. (2018).

IMM: Representation, Expert Guidance, and RL

IMM combines predictive representation learning, expert guidance, and continuous-control RL.



What SRLU + TCSA means. SRLU first encodes local limit-order-book patterns; TCSA is applied on top of these SRLU states and selects the most informative time positions and channels for the current decision.

Adapted from Niu et al. (2023), IMM, Figure 3.

Predictive Representation: Information Across Horizons

The encoder is trained to preserve market information useful at several future horizons.

$$\mathcal{H} = \{10, 60, 120, 300\} \text{ s}, \quad \hat{y}_{t,h} = g_h(f_\psi(H_t))$$

Auxiliary training loss

$$L_{\text{pred}}(\psi) = \sum_{h \in \mathcal{H}} \omega_h \mathbb{E}[\ell(\hat{y}_{t,h}, y_{t,h})].$$

Qualification Future data may define training targets $y_{t,h}$, but only time- t information may enter the deployed policy state.

Niu et al. (2023), IMM. Forecast horizons refer to the current research setup.

TD3 Part 1: Why Twin Critics Are Needed

TD3 is the continuous-action backbone: it learns from replay, but explicitly fights Q-value overestimation.

Off-policy critic target

Critic regression

$$\tilde{a}' = \text{clip}_{\mathcal{A}}(\pi_{\theta'}(s') + \epsilon), \quad \epsilon \sim \text{clip}(\mathcal{N}(0, \sigma^2)).$$

$$y = r + \gamma(1 - d) \min_{j=1,2} Q_{\phi'_j}(s', \tilde{a}').$$

$$L_Q(\phi_i) = \mathbb{E}_{\mathcal{D}} (Q_{\phi_i}(s, a) - y)^2, \\ i = 1, 2.$$

Interpretation. The minimum of two target critics is a conservative target: it reduces the tendency to exploit accidentally overestimated actions.

Key point In market making this matters because a tiny optimistic error can push the policy toward aggressive quote sizes or spreads.

TD3 Part 2: Actor Updates and Stabilization

TD3 improves the actor more slowly than the critics, so the policy follows a more reliable value estimate.

Delayed policy update

$$\max_{\theta} \mathbb{E}_{s \sim \mathcal{D}} Q_{\phi_1}(s, \pi_{\theta}(s)),$$
$$\theta \leftarrow \theta \quad \text{only every } d \text{ critic steps.}$$

Soft target networks

$$\phi'_i \leftarrow \tau \phi_i + (1 - \tau) \phi'_i,$$
$$\theta' \leftarrow \tau \theta + (1 - \tau) \theta'.$$

Mechanism	Market-making interpretation
Twin critics	do not trust a quote only because one critic is optimistic
Target smoothing	avoid actions that work only in a tiny neighborhood
Delayed actor	update the policy after critic estimates become less noisy
Soft targets	keep Bellman targets stable across replay updates

Fujimoto, van Hoof and Meger (2018), TD3; Niu et al. (2023), IMM.

Imitation Loss: Expert Guidance During Policy Learning

Expert actions provide a prior for the actor while reinforcement learning remains the optimization target.

$$a^E = \pi_E(S), \quad L_{\text{BC}}(\theta) = \mathbb{E}_{(S, a^E) \sim \mathcal{D}_E} \|\pi_\theta(S) - a^E\|_2^2.$$
$$L_{\text{actor}}^{\text{IMM}} = L_{\text{RL}} + \beta_t L_{\text{BC}}, \quad \beta_{t+1} \leq \beta_t, \quad \beta_t \geq 0.$$

Qualification The expert guides early learning; it does not guarantee optimal actions or final policy quality.

Niu et al. (2023), IMM. Equations show the conceptual imitation objective.

Selected References: Market Making and IMM

Market microstructure

Gould, M. D., et al. (2013). *Limit order books*. Quantitative Finance, 13(11), 1709–1742.

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